

Department of Electrical and Electronics Engineering

University of Dhaka
Batch - 103, EEE-58

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Course: EEE-2109

Title: Signals and Systems

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Note by
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SYLLABUS

Signals: Classification of Signals, Basic Operations on Signals, Elementary Signals, The Concept of Frequency in Continuous-Time and Discrete-Time Signals, Analog-to-Digital and Digital-to-Analog Conversion.

Systems: Input-Output Description of Systems, Classification of Systems, Properties of Systems, Interconnection of Systems.

Linear Time-Invariant (LTI) Systems: Response of Continuous-Time LTI System and Convolution Integral, Response of a Discrete-Time LTI System and Convolution Sum, Recursive and Non-Recursive Systems, Differential and Difference Equation Representations of LTI Systems, Solution of Differential and Difference Equations, State Variable Descriptions of LTI Systems, Structures for the Realization of LTI Systems.

Fourier Analysis of Signals and Systems: Fourier Representations of Signals, The Fourier series, The Discrete-Time Fourier Series, The Fourier Transform, The Discrete-Time Fourier Transform, Properties of Fourier Representations, Parseval's Theorem, Power Density Spectrum and Energy Density Spectrum.

Applications of Fourier Analysis: Frequency Response of LTI Systems to Complex Exponential Signals, Filtering and Bandwidth, Frequency Analysis Analog Electrical Systems, Sampling and Reconstruction of Signal, Convolution and Multiplication with Mixtures of Periodic and Non-periodic Signals, Modulation and Demodulation, Time-Division and Frequency-Division Multiplexing.

The Laplace Transform: Definition, Properties, Convergence of Laplace Transform, Initial-value and Final-Value Theorem, Inverse Laplace Transform, Partial-fraction Expansions, Heaviside Expansion Formula, Solution of Differential Equation, System Transfer Function, System Stability, Electrical Network Analysis.

REFERENCE BOOKS

1. Signals and Systems, 2^{nd} Edition, Simon Haykin and Barry Van Veen, Wiley Inc.
2. Signals and Systems, 2^{nd} Edition, Alan V. Oppenheim, Alan S. Willsky and S. Hamid, Pearson Education.
3. Linear Systems and Signals, 3^{rd} Edition, B.P. Lathi and Roger Green, Oxford University Press.
4. Signals and Systems, 1^{st} Edition, Sanjit K. Mitra, Oxford University Press
5. Continuous and Discrete-Time Signals and Systems, 2^{nd} Edition, Samir S. Soliman and Mandyam D. Srinath, Prentice-Hall of India.

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CHAPTER 1

SIGNALS

1.1 Introduction

Signal : Signal is a function of one or more independent variables that carries information (usually through amplitude, phase, frequency and position etc.) about the behavior of the physical variable/phenomenon.

$$F(x_1, x_2, x_3, \dots, x_n)$$

Here,

$F \rightarrow$ Signal

$x_1, x_2, x_3, \dots, x_n \rightarrow$ independent variable

- Anything which is constant is not a signal.
- AC $\rightarrow$ Signal, DC $\rightarrow$ Not a signal.

Types of Signal :

There are two types of signal.

- (i) Single variable signal : Depends on one independent variable.
Example : $f(x), g(t)$
- (ii) Multi-variable signal : Depends on more than one independent variable.
Example : $f(x_1, x_2), g(t_1, t_2, t_3)$

System : The meaningfull interaction of physical devices and components is called as system.

- System alone can't achieve anything, it must link with signal.

$$i/p \longrightarrow \boxed{\text{System}} \longrightarrow o/p$$

- i/p signal $\longrightarrow f(x_1, x_2)$
- o/p signal $\longrightarrow g(x_1, x_2)$ Both are same independent variable.

For Analysis problem,

i/p signal and system are given. Find out o/p signal.

For Synthesis problem,

i/p signal and o/p are given. Find out system of the signal.

1.2 Continuous and Discrete Time Signals

Continuous Time Signal(CTS) : Specified for every value of time t .

Example

Discrete Time Signal (DTS) : Specified at discrete time intervals.

Example:

Uniform Discrete Time Signal : where every time interval is same.

$$\Delta t = T$$

Example :

1.2.1 Addition of CTS

1.2.2 Multiplication of CTS

1.3 Scaling of CTS

There are two types of scaling in CT signal.

1. Time scaling.
2. Amplitude scaling.

1.Time scaling : The compression or expansion of a signal in time.

$$x(t) \xrightarrow{T.Sc} y(t) = x(\alpha t)$$

Case-1

$$|\alpha| > 1 \longrightarrow \alpha \in (-\infty, -1] \cup [1, \infty)$$

This is the case of compression.

Case-2

$$|\alpha| < 1 \longrightarrow \alpha \neq 0 \quad [\alpha = -0.3, -0.2, -0.1, -0.099 \dots \dots 0.1, 0.2, 0.3]$$

This is the case of expansion.

Shortcut

- Amplitude remain same.
- Times are divided by α .

2.Amplitude Scaling : The increased and decreased of a signal in time.

$$x(t) \xrightarrow{A.Sc} y(t) = \beta x(t)$$

Case-1

$$|\beta| > 1 \longrightarrow \beta \in (-\infty, -1] \cup [1, \infty)$$

This is the case of amplitude expansion.

Case-2

$$|\beta| < 1 \longrightarrow \beta \in (-1, 0) \cup (0, 1)$$

This is the case of amplitude reduction.

1.4 Shifting of CTS

There are two types of shifting in CT signal.

1. Time shifting.
2. Amplitude shifting.

1.Time Shifting : Time shifting means shifting a signal in time, either forward (delay) or backward (advance), without changing its shape.

$$x(t) \xrightarrow{T.Sf} y(t) = x(t + K)$$

Case-1 $k > 0$

This is the case of left shifting or time advanced.

Case-2 $K < 0$

This is the case of right shifting or time delay.

Advancing time signal is not possible in real life.

2.Amplitude Shifting : Amplitude shifting means adding or subtracting a constant value to the signal's amplitude, which shifts it vertically up or down on the graph.

$$x(t) \xrightarrow{A.Sf} y(t) = x(t) + l$$

Case-1 $l > 0$

This is the case of upward shifting.

Case-2 $l < 0$

This is the case of downward shifting.

1.5 Folding/Reversal of CTS

1.Time Reversal : Time reversal means flipping a signal across the vertical axis ($t = 0$ line).It inverse signal in time.

$$x(t) \xrightarrow{T.Sc} y(t) = x(\alpha t)$$

$$x(t) \xrightarrow[\alpha=-1]{T.R} y(t) = x(-t)$$

This is the special case of time scaling with $\alpha = -1$.It is also called as folding or reflection.

2.Amplitude Reversal : Amplitude reversal means flipping a signal across the horizontal axis (amplitude axis). It changes the sign of the amplitude values.

$$x(t) \xrightarrow{A.Sc} y(t) = \beta x(t)$$

$$x(t) \xrightarrow[\beta=-1]{A.R} y(t) = -x(t)$$

This is the special case of amplitude scaling with $\beta = -1$.It is fold signal with respect to Y-axis.

1.6 Multiple Transformation Of CTS

$$x(t) \longrightarrow \boxed{\text{System}} \longrightarrow y(t)$$

Problem 1. Draw the signal $y(t)$